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SEG Automotive Germany GmbH
Sophia Neubauer, Human Resources
Lotterbergstraße 30, 70499 Stuttgart

Stuttgart, 21 October 2026

Application: Working Student (m/f/d) Hardware Architecture for Automotive Electronics

Dear Sophia Neubauer,

Automotive electronics depends on hardware architectures with clear interfaces, documented concepts and requirements that can be traced through the whole product lifecycle. Your Working Student role in Hardware Architecture, which combines architecture models with functional safety according to ISO 26262 on real customer projects, is exactly the step I want to take. I am an M.Sc. Electrical Engineering student at the University of Stuttgart with coursework in power systems, power electronics and battery modeling.

The posting asks for a basic understanding of electronic hardware, circuit design and system-level engineering. During a one-year internship at FocalSystech, I defined electrical and hardware–software interfaces between sensors, actuators and microcontrollers (CAN, I2C, SPI, UART) and supported their system integration. I designed and validated circuits, evaluated and selected components against technical requirements, and created PCB layouts in Altium. Defining an interface so that independent parts work together reliably is, to me, the core of hardware architecture.

In a university team project on battery cell modeling, I analysed how electrical, thermal, aging and manufacturing tolerances affect cell behaviour and built that analysis into a simulation model. It taught me to represent a system in a structured, checkable way, which I expect to carry over to architectural models, and to work carefully and in detail.

I want to be clear about what I have not done yet. I have not worked with Enterprise Architect or Cameo Systems Modeler, I have used MATLAB but not Simulink, and I have not applied ISO 26262 in practice. I am motivated to learn all three, and I see this role as the right place to learn functional safety alongside hardware, system and safety engineers in an international team. My English is fluent. My German is at A2 and I am preparing for the telc B1 exam. I am glad your posting treats German as a plus.

I can work up to 20 hours per week alongside my studies, with a start date by agreement, and I am comfortable with your hybrid model at the Weilimdorf site. I would welcome the chance to discuss how I can support your hardware architecture team.

Kind regards,

Vineeth Raghav Gururajan

Enclosures: CV, proof of enrolment, certificates